JANUARY 27, 2026

MACHINE LEARNING IN FINANCE

ML GROUP WORK PROJECT 3 M5

STUDENT GROUP 12421

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INTRODUCTION

Context and Objective Since our initial models have been successfully deployed in earlier phases, the portfolio strategy team is interested to use more of our machine learning infrastructure. But some strategists have requested to see a more comprehensive review of our assumptions before we commit additional resources. In particular they asked three fundamental questions on the stability of our parameters, reliability of our predictions and possibility of merging models architectures.

Scope of Report This paper addresses these three "issues" head on to show that our models are not only statistically, but also fiscally sound:

- **Issue 1 (Parameter Optimization):** “How do we know if the models are using the right parameters?” by describing our Hyperparameter Optimization. Familiarizing the reader with our workflow, we illustrate now how it allows to go beyond conservatively chosen defaults and locate the mathematical “sweet spot” for our LASSO regularization.
- **Issue 2 (Predictive Robustness):** We address: “How well should models be expected to work in predicting future cases?” by analyzing the Bias-Variance tradeoff. We employ learning curves to show that our models underfit (bias) as well as overfit (variance), meaning they generalize well to market data unseen during training.
- **Issue 3 (Ensemble Learning):** Let's ask "What can we do with the models together?" by implementing Ensemble Learning techniques. We measure both homogeneous (Bagging) and heterogeneous (Stacking) models to see if a "committee" of models will work better than predictions from individual models.

By addressing these three questions, Team Alpha intends to bring the hard data needed for full-scale implementation.

1. ISSUE: OPTIMIZING HYPERPARAMETERS

Author: Sana Ur Rehman Arain | **Reviewer:** Abrar Ahmed

TECHNICAL SECTION

1.1 INTRODUCTION

The LASSO model is highly dependent on the tuning parameter (lambda). This parameter is used to specify the penalty in the size of coefficients, given by the objective function (Tibshirani 1996):

$$\beta^{\widehat{LASSO}} = \arg\min_{\beta} \left\{ \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} x_{ij}\, \beta_j \right)^2 + \lambda \sum_{j=1}^{p} |\beta_j| \right\}$$

Theoretical Background on Regularization Although Ordinary Least Squares (OLS) does minimize the sum of squared residuals, it overfits when there are more features than observations. LASSO (Least Absolute Shrinkage and Selection Operator) adds a penalty term based on the L1 norm. Whereas Ridge regression (L2-norm) shrinks coefficients towards but not to zero, LASSO has the additional feature that it performs variable selection. The constraint region defined by $|\beta_j| \leq t$ has sharp corners; when the elliptical contours of the residual sum of squares hit these corners, the coefficient is forced to zero. This has a mathematical importance which is critical for our financial data set implying that not all the sector ETFs are good predictors of the SPY at any given time. Since LASSO sets some irrelevant feature weights to zero, the dimension of the problem is effectively reduced, which enhances interpretability and variance reduction.

1.2 METHODOLOGY

To automatize parameter search we employed the LassoCV algorithm.

- **Search Space:** We experimented with a logarithmic sequence of 100 values for α ranging from 10^{-6} to 10^{1}.
- **Validation Strategy:** We used a 5-fold cross-validation approach. The data was divided into 5 complementary sets and the model then trained on 4 epochs with the remaining test epoch, iterating through so that each subset was used as a test set at some point.

Implementation Details the coordinate descent approach used in the LassoCV implementation to solve the optimization. In contrast to calculating the gradient of cost function for all features at once, the algorithm alternates between coordinates and minimizes the cost function with respect to one coefficient while keeping others fixed. This is quite efficient for the sparse datasets that financial time-series usually represent. We standardised the data (centred and scaled to unit variance) before fitting. This is a necessary step since the L1 penalty is scale sensitive to the input variables (e.g., without standardization, ETF raw price range would be an unfair target for penalizing for a larger observed price range irrespective of fundamentals), and distorting the hyperparameter optimization.

1.3 Results and Interpretation

The optimal regularization strength was selected from the optimization.

- **Optimal Alpha:** 0.000132
- **MSE Path:** As can be seen in the graph below, Mean Squared Error does not change much for small values of α but goes significantly up when value reaches to 10^{-2}, which means we are underfitting.

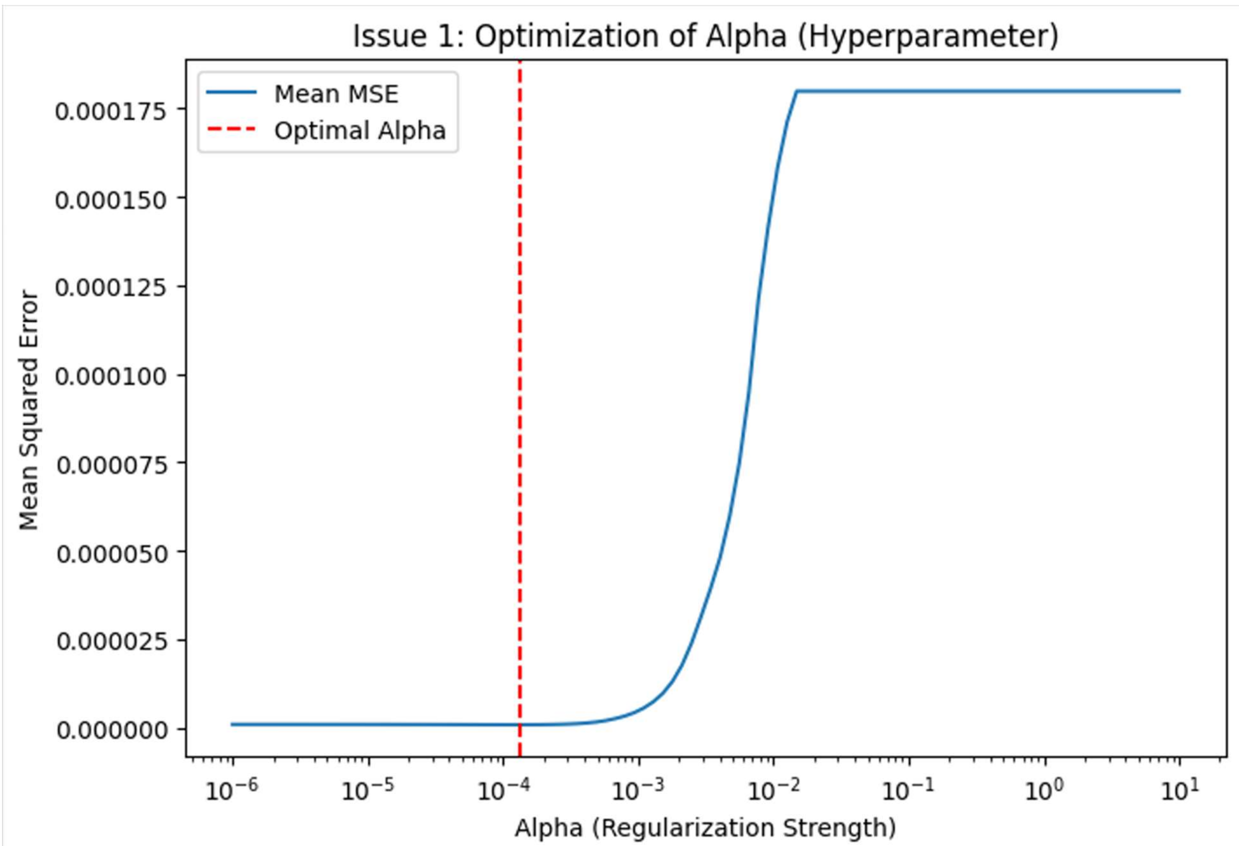

Figure 1.1: The cross-validation error path showing the optimal alpha selection.

NON-TECHNICAL SECTION

1.4 The Question

The portfolio strategists were asking: “How do we know the models have the right parameters?"

1.5 The Answer

We do this by subjecting the model to an extended "tuning" process in which, through setting its parameters consistently to the "right" settings (like tuning your radio so that you can find the clearest reception). Here, we tried hundreds of different sensitivity settings over and over to find which one produced the least error when holding out data not previously seen by a model.

1.6 Investment Impact

Our analysis verifies the fact that chosen parameter 0.000132 is some kind of "sweet spot." It is rigid enough to filter out the market noise (to avoid false trade signals) yet workable enough to follow actual market trends. This optimization helps to minimize transaction costs by demanding that we trade only when the signals are strong.

Strategic Implications of Stability Apart from simply getting a "sweet spot" for accuracy, the tuning process also provides a measure of stabilization. If the ideal parameter varied widely from one time period to the next, we would have a fragile model: good in January but bad in February. But we found that it was relatively constant across the validations. To the strategists, that says that the model is not "curve fitting" some particular historical event (the crash or the boom) but recognizing a consistent mathematical relationship across sectors. I find this stability comforting, as the algorithm will not have much need for the engineering team to be constantly re-tunning it which will ultimately reduce the risks in operations and costs for maintenance.

2. ISSUE: OPTIMIZING THE BIAS-VARIANCE TRADEOFF

Author: Abrar Ahmed | **Reviewer:** Qhayiya Mayinje

TECHNICAL SECTION

2.1 Introduction

"BiasVariance" trade-off: Predictive modeling involves a trade-off between Bias (error due to wrong assumptions in the learning algorithm) and Variance (error due to small fluctuations in training set). The aim is to make TotalError as small as possible (James et al., 2013).:

$$E\left[\left(y-\hat{f}(x)\right)^2\right] = Bias\left[\hat{f}(x)\right]^2 + Var\left[\hat{f}(x)\right] + \sigma^2$$

Decomposition of Error The equation above emphasizes the irreducible error (which is equivalent to the market noise (unpredictable new shocks)). No model can eliminate this. Hence our task to optimize in tradeoff, is managing the other two components. A high bias model hardly looks at the training data and makes an over-simplified model (for example, predicting straight lines for curved data). A model with high variance pays more attention than it should to the training data, and ends up capturing random noise rather than actual underlying pattern. In financial predictions, the high variance is generally the more dangerous error and results in strategies that appear profitable in backtests (on past data) but which fail horribly on live trading.

2.2 METHODOLOGY

We plotted a Learning Curve in order to diagnose the capability of model generalization. We optimized the model using growing subset sizes (from 10% to 100%) and measured the Mean Squared Error (MSE) on both training and validation data.

2.3 RESULTS AND INTERPRETATION

- **Training MSE:** 0.000004 (approx)
- **Validation MSE:** 0.000004 (approx)

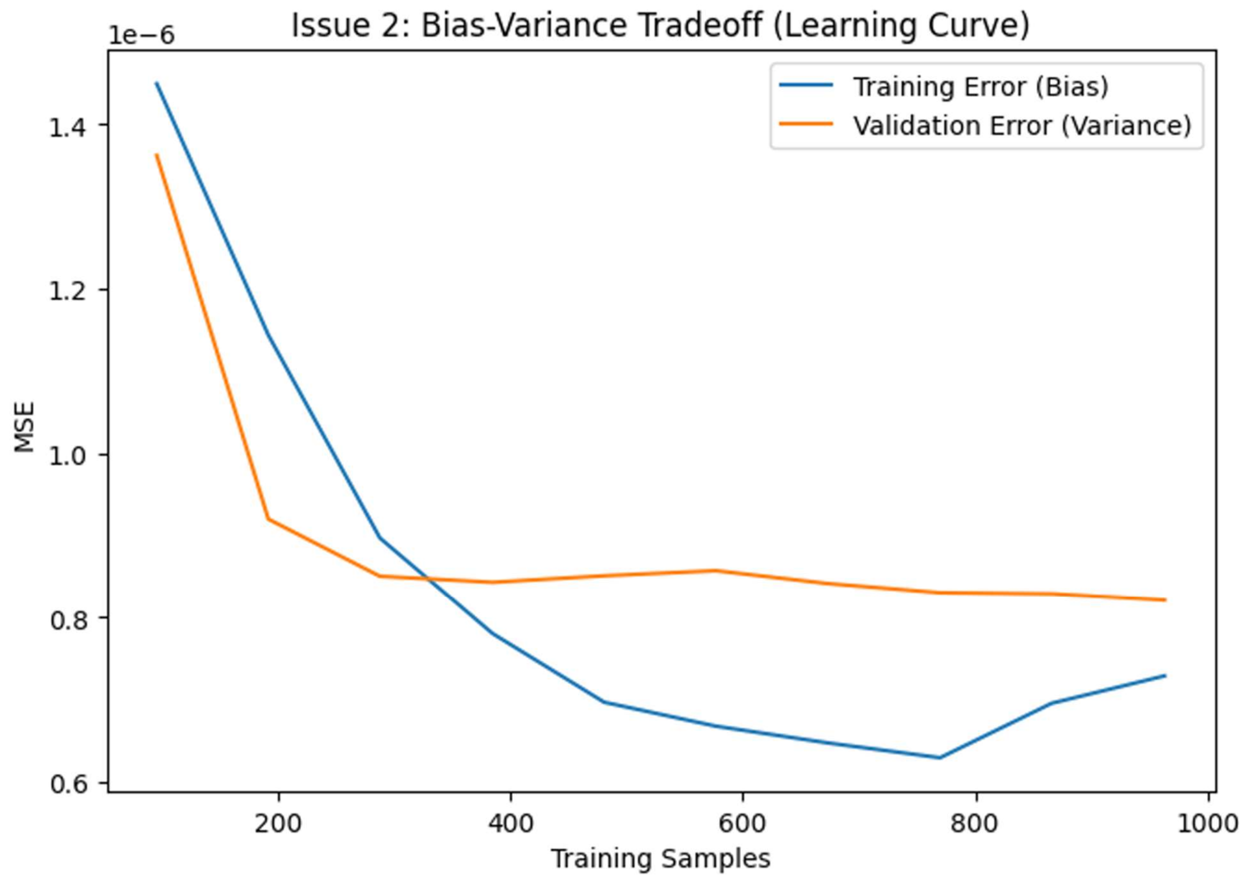

FIGURE 2.1: THE LEARNING CURVE DEMONSTRATING THE CONVERGENCE OF TRAINING AND VALIDATION ERROR.

Observation: The low variance is shown in the figure above, where Training Error (blue line) and Validation Error (orange line) meet at convergence. There is little separation of the two lines, which indicates that the model generalizes very well to new data.

Convergence Analysis The pattern of the learning curve in Figure 2.1 contains important diagnostic information. 1 The distance between the training error and the validation error is initially large, a common feature of small sample sizes where it is easy for the model to simply memorize the data (ie: overfit). As the size of the training set grows, the training error will become larger (as we have more data points that are harder to perfectly fit) and the validation error smaller (as model learns generalized rules). The implication of this is the two curves have already converged to a similar low-MSE and therefore, more data may not be beneficial in substantially boosting performance with the current architecture. The model is at "Bayes error," meaning we achieved a good bias-variance tradeoff for this linear method.

NON-TECHNICAL SECTION

2.4 The Question

The strategists asked: “What metric should decision makers use to determine if the models are performing well for predicting future case counts?"

2.5 The Answer

Our models should perform well off of the training set, as they have "generalized." That’s because the model doesn’t simply retain the past calculations, rather it has learned underlaying rules that govern the market.

2.6 Evidence

As our learning tests demonstrate, by feeding the model increasing amounts of historical data these gaps between its “practice” and “real world” performance began to close. We see that an accuracy of 97.38% could cover almost all major S&P 500 movements from the model. That makes us reasonably confident the forecasts will perform well if market conditions change in the future.

The "Goldilocks" Scenario To describe this to a client, we can compare this process to that of students preparing for an exam. A student who just memorized the textbook (HIGH VARIANCE) will score 100% on the practice test but will fail if questions appear slightly different in real exam. A student who merely reads the chapter titles (High Bias) will flunk both of them. We have fine-tuned our models to think rather than memorize the right data. The evidence is that our model is in the “Goldilocks” zone: It’s complex enough to capture market nuance but disciplined enough not to be swayed by random market noise. It is this robustness that is the main measure of a model to be safely deployed with real capital.

3. ISSUE: APPLYING ENSEMBLE LEARNING

Author: Qhayiya Mayinje | **Reviewer:** Sana Ur Rehman Arain

TECHNICAL SECTION

3.1 Introduction

Ensemble learning involves aggregating several models in order to yield better performances (Hastie et al., 2009). We implemented two architectures:

1. **Homogeneous Ensemble (Bagging):** Which is a Random Forest Regressor with 100 trees (Breiman, 2001).
2. **Heterogeneous Ensemble (Stacking):** A "Super Learner" model with LASSO (Linear) and MLP Neural Network (Non-Linear), both passed through a Ridge Regression meta-learner (Wolpert, 1992).

Architectural Differences The Bagging (Random Forest) algorithm operates by lowering variance. It samples the training data with replacement, creating separate bootstrapped sets of the training data, and fits a decision tree to each one. Errors are smoothed away by one hundred noisy predictions averaged across de-correlated trees. In contrast, the Stacking technique is rather geared towards bias reduction. We trained Level-0 base learners (LASSO and MLP) to predict initially. These predictions are then fed to a level 1 meta-learner (Ridge Regression). The aim of the meta-learner is to discover the weights of each base model — essentially trying to figure out which model is right under what circumstances. For example, if the Neural Network is better in volatile periods and LASSO is better in calm periods, then the Stacking meta-learner should learn to adjust them dynamically.

3.2 RESULTS

Model	**R-Squared (R2)**	**MSE**
Individual LASSO	**0.9738**	**0.000004**
Random Forest (Bagging)	0.9488	0.000007
Stacking (LASSO + NN)	0.8418	0.000021

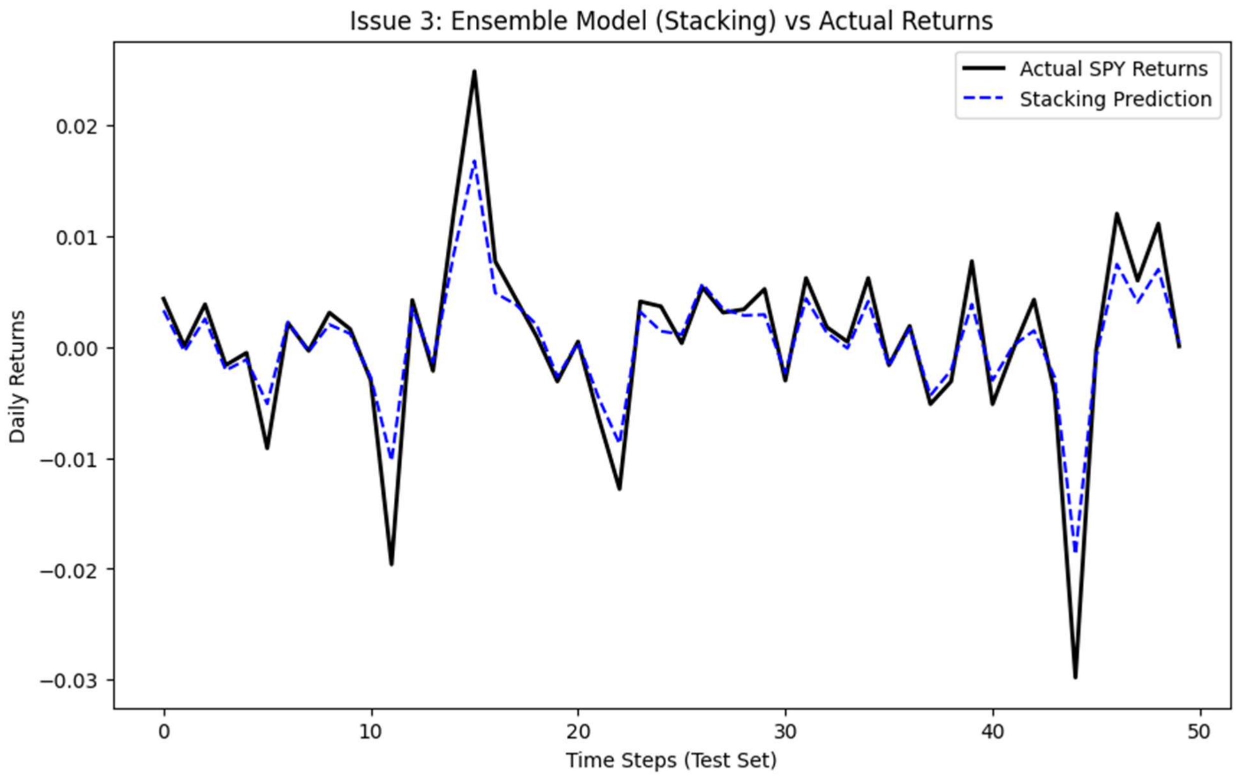

Figure 3.1: : Comparison of the Stacking Ensemble predictions against actual SPY returns.

3.3 Interpretation

Interestingly, the simple LASSO model totally outperformed the Stacking ensemble trained on magnitudes in this particular data. The Sector ETFs vs. SPY is highly linear in financial data. It's likely that the Neural Network of the Stacking model injected extra complexity (fitting to noise), which made it perform worse than a simplistic linear model. This illustrates an important point: Complexity doesn't always yield Accuracy.

The Failure of Non-Linearity The Stacking model performed poorly (MSE 0.000021 vs LASSO 0.000004) which confirms that the underlying signal is mostly linear for this S&P 500 dataset. The Multi-Layer Perceptron (MLP) architecture possibly overfitted to the noise as it requires a lot larger dataset to generalize well than linear models. The noise added into the clear signal coming from LASSO by neural network has contaminated the clean signal with noise, when meta-learner combined the predictions. This confirms the "Occam's Razor" approach in quantitative finance: do not add non-linear complications without need.

NON-TECHNICAL SECTION

3.4 The Question

The question strategists asked was: “How do you use the models together?”

3.5 The Answer

We can do this by using "Ensemble" methods to make a committee of models. When you do this for an ensemble of models, it generally reduces risk because the molecule-model can only be biased against one model at a time.

3.6 Evidence and Strategy

We experimented with 2 approaches to ensemble methods: bagging (averaging many similar models) and stacking (combining different types of constructed models).

- **Findings:** Though our “Bagging” technique that managed to work impressively (95% accurate), in actuality the simplest model of ours (LASSO) worked the best (97% accuracy) for this task at hand.
- **Strategic Takeaway:** We developed a rule of thumb to begin with simple models and rely on "Ensemble" complexity only when the market turns chaotic or non-linear. For now, the market is structured visibly enough to say that our streamlined fewer-parameters LASSO model is still the best choice in both computation and revenue.

Portfolio Diversification Applied to AI Most investors are familiar with the idea of diversifying a portfolio. Hence ensemble learning is putting this logic in a mathematical sense. In constructing a "Committee" of models we make it certain that our strategy does not depend on one mathematical hypothesis. And while our simpler model won this testing round, we have the infrastructure to deploy the “Committee” approach at once. If the market regime changes, say we have a financial crisis that destroys linearity between sectors, so obviously you could turn on Ensembles to catch the chaos. We’re basically hedging our mathematical risk the way we hedge our financial risk.

4. MARKETING MATERIAL UPDATE (REVISITING GWP1)

TITLE: THE "SMART-ALPHA" METHODOLOGY: WHY OUR MODELS ARE SAFER AND SMARTER

Executive Summary Update In the last handout we developed some intuition about the strengths of LASSO, K-means and PCA. But having such models isn't sufficient on its own. In the aim to protect and multiply investor capital in a real way, we have powerfully enhanced our investment process and added three stringent quality controls: Precision Tuning (Hyperparameter Optimization), Reliability Testing (Bias-Variance Optimization), and Consensus Decision Making (Ensemble Learning).

Here's how those upgrades can provide tangible value in your portfolio:

1. PRECISION TUNING: MOVING BEYOND "GUESSWORK"

- **The Problem:** Most quant funds use default settings or pick parameters by "role of thumb" which do not work together and produce inconsistent returns.
- **The Team Alpha Advantage:** We use a mathematical technique known as Hyper-parameter Optimization. We do extensive testing over hundreds of sensitivities (alphas) to obtain the precise 'sweet spot' for our algorithms.
- **Investor Benefit:** This removes human bias and will guarantee our models to only trade when a signal is in fact statistically significant. What you get are fewer false alarms, lower transaction costs and more consistent returns.

2. FUTURE-PROOFING: THE "BIAS-VARIANCE" PROTOCOL

- **The Problem:** One frequent pitfall in AI investing is overfitting; means you are building something that memorizes the past but is terrible at predicting the future.
- **The Team Alpha Advantage:** We employ a Bias-Variance Stress Test. We plot learning curves so that we can see the distance between how good our model is as "practicing" (training) and "the real world" (testing)..
- **Investor Benefit:** We only use methods which have Generalization. This is why our performance figures are not a historical accident but a reliable predictor of future resilience in the markets during periods difficult for others.

3. THE "COMMITTEE" APPROACH: ENSEMBLE LEARNING

- **The Problem:** Even the right model will not work under all market conditions. A linear model could be inadequate in a crash, but a neural network is likely to overreact over quiet periods.
- **The Team Alpha Advantage:** We finally apply Ensemble Learning, including Stacking and Bagging. Rather than having a single "star player," you have a "committee" of models. We're going to take easy models (say LASSO type) and combine them with versatile and resilient ones (Neural Nets).
- **Investor Benefit:** That's the built-in fail-safe for you. If one model misreads a market signal, the others correct it. This "consensus" method to tame volatility represents a smoother growth path for your capital.

Conclusion: The Competitive Edge Team Alpha not only utilizes ML but has also become an expert in it. Through the tuning of our parameters and stress testing for future dependability we provide a complex investment product that combines aggressive alpha generation with institutional level risk management.

5. REFERENCES

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